

Multiple Access

Difference Between Multiplexing & Multiple access:

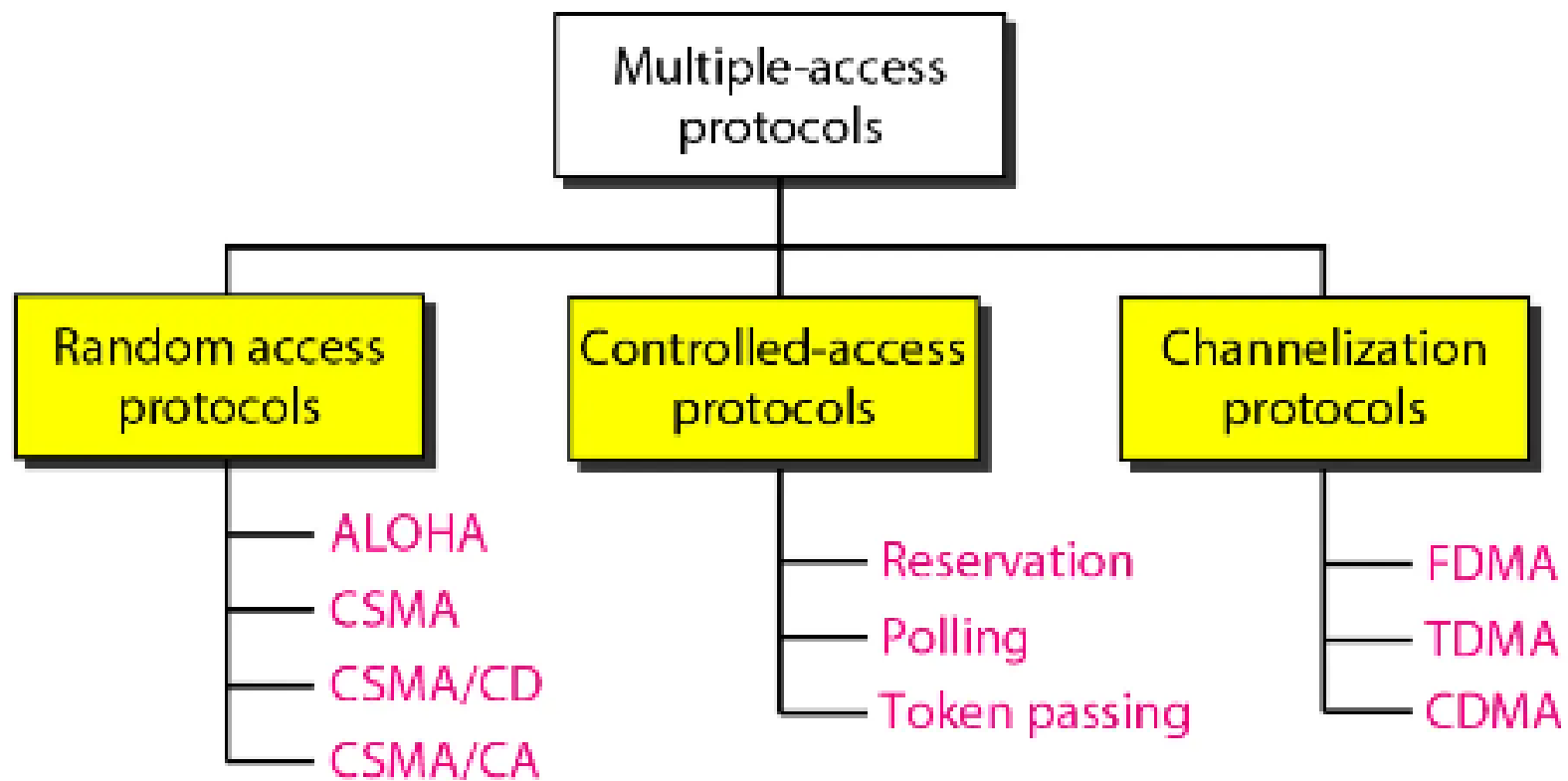
Multiplexing combines **multiple signals** from one source into a single channel, while multiple access allows **several users** to share a single channel.

1. Multiplexing

- **Goal:** Combine **multiple signals** → send through **one channel**.
- **Used in:** Communication **between two points** (point-to-point).
- **Focus:** Efficient use of **channel bandwidth**.
- **Examples:**
 - **TDM** (Time Division Multiplexing)
 - **FDM** (Frequency Division Multiplexing)
 - **WDM** (Wavelength Division Multiplexing)
- **Analogy:**
Like several TV channels sharing one cable using different frequencies.

2. Multiple Access

- **Goal:** Allow **multiple users** to access the same medium **at the same time**.
- **Used in:** Communication in **networks**, where many users share the channel.
- **Focus:** Efficient sharing of **a common medium**.
- **Examples:**
 - **TDMA** (Time Division Multiple Access)
 - **FDMA** (Frequency Division Multiple Access)
 - **CDMA** (Code Division Multiple Access)
 - **OFDMA** (Orthogonal Frequency Division Multiple Access)
- **Analogy:**
Many people speaking through different “slots” or “codes” on the same network.



*In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.*

Topics discussed in this section:

ALOHA

Carrier Sense Multiple Access

Carrier Sense Multiple Access with Collision Detection

Carrier Sense Multiple Access with Collision Avoidance

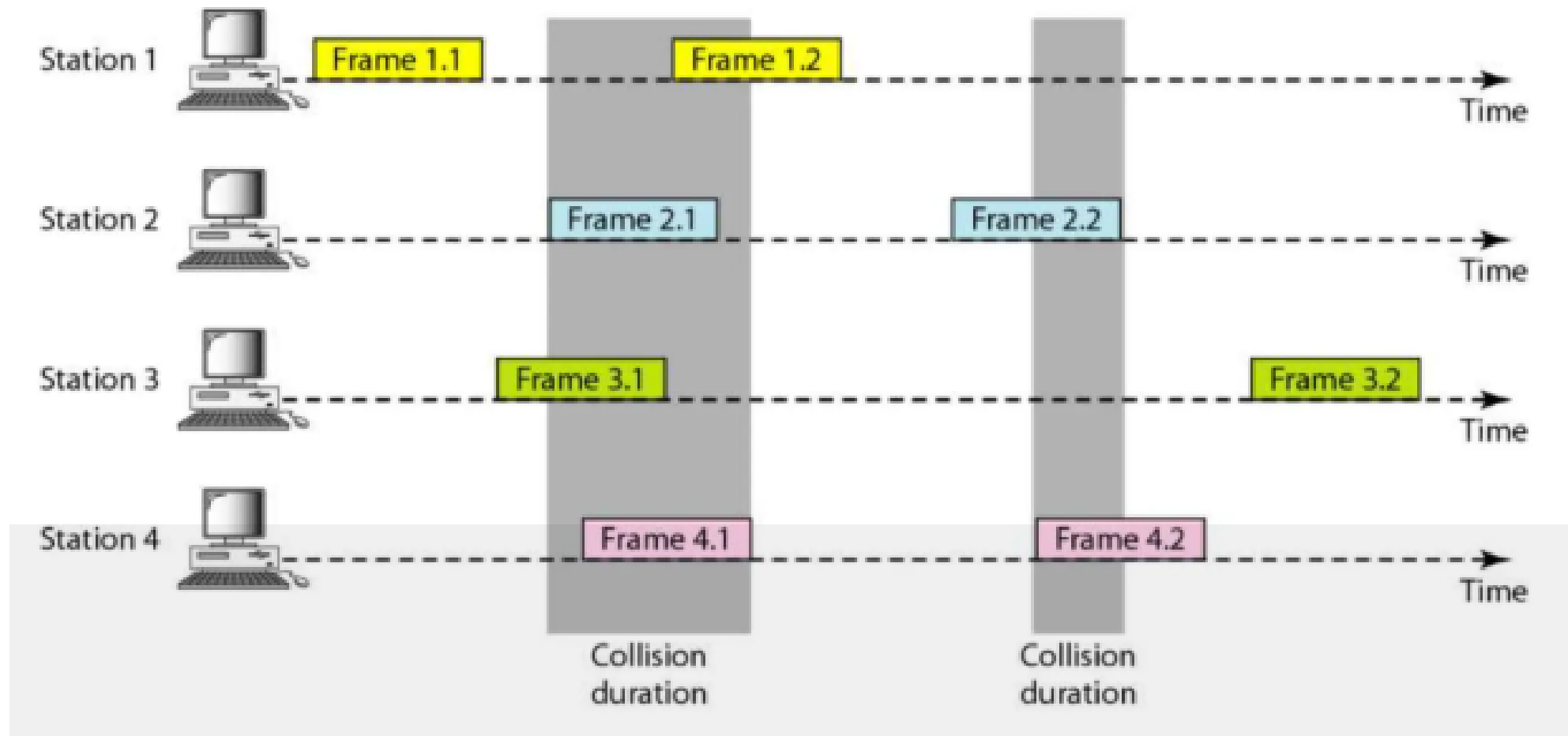
RANDOM ACCESS PROTOCOL

- ALOHA
(Pure ALOHA and Slotted ALOHA)
- CSMA
- CSMA/CD
- CSMA/CA

ALOHA Protocol

- The original ALOHA protocol is called *pure ALOHA*.
- Each station sends a frame whenever it has a frame to send.
- However , there is only one channel for them to share.

Figure 12.3 *Frames in a pure ALOHA network*



SLOTTED ALOHA

- *In slotted aloha we divide the time into slots.*
- *Frames can only be transmitted at beginning of slot: "discrete" ALOHA.*

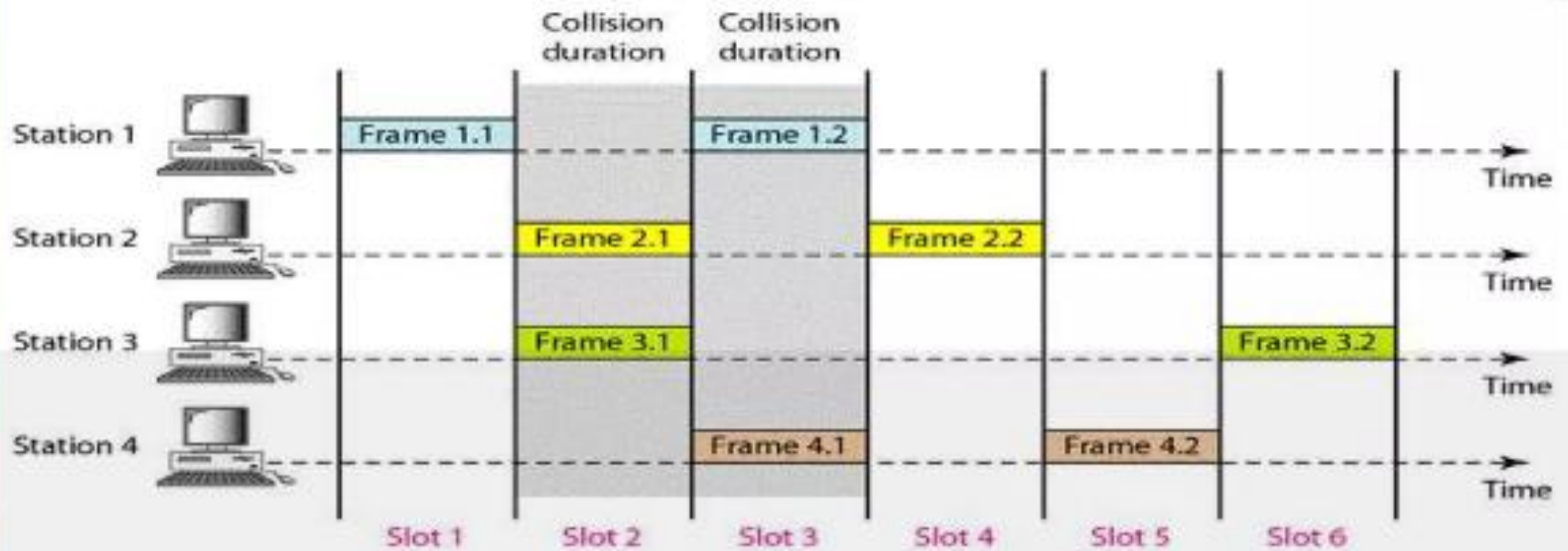
Slotted ALOHA

we assume,

- All frames consist of exactly L bits.
- Time is divided into slots of size L/R seconds
- Nodes start to transmit frames only at the beginning of slots
- The nodes are synchronized so that each node knows when the slots begin.
- If two or more frames collide in a slot ; then all the nodes detect the collision event before the slot ends



SLOTTED ALOHA



CARRIER SENSE MULTIPLE ACCESS

- CSMA protocol was developed to overcome the problem found in ALOHA i.e. to minimize the chances of collision, so as to improve the performance.
- CSMA protocol is based on the principle of 'carrier sense'.
- The chances of collision can be reduce to great extent if a station senses the channel before trying to use it.
- Although CSMA can reduce the possibility of collision, but it cannot eliminate it completely.

*In **controlled access**, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.*

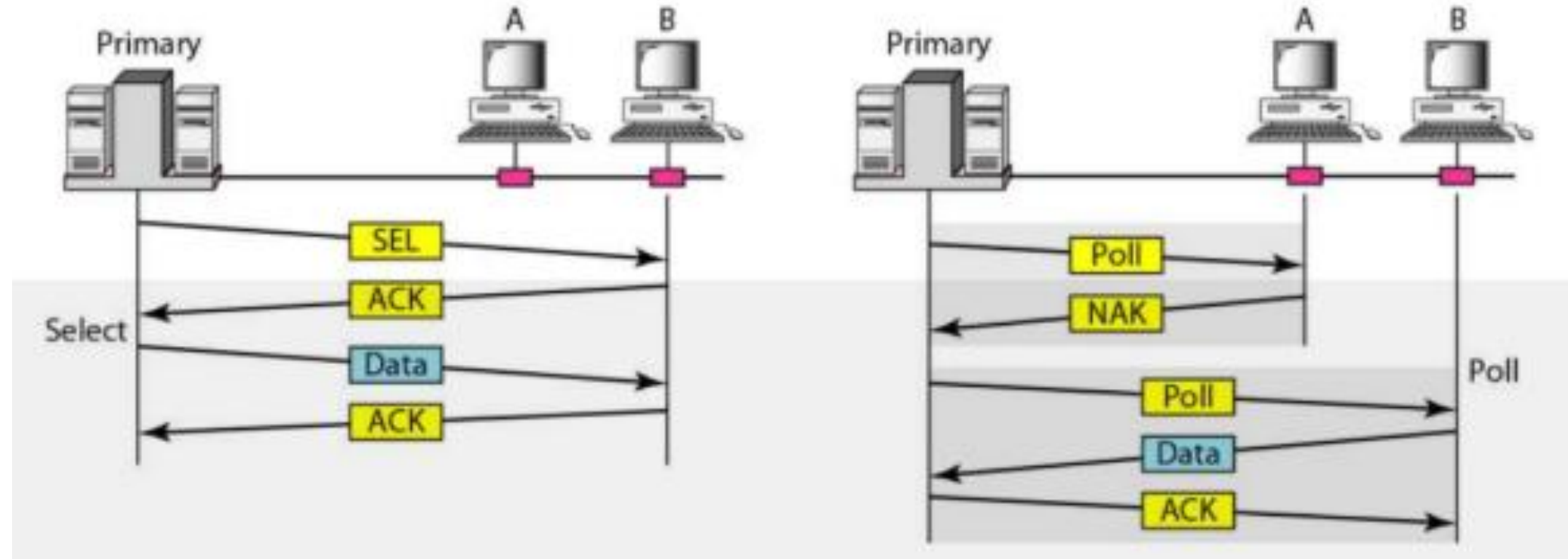
Topics discussed in this section:

Reservation

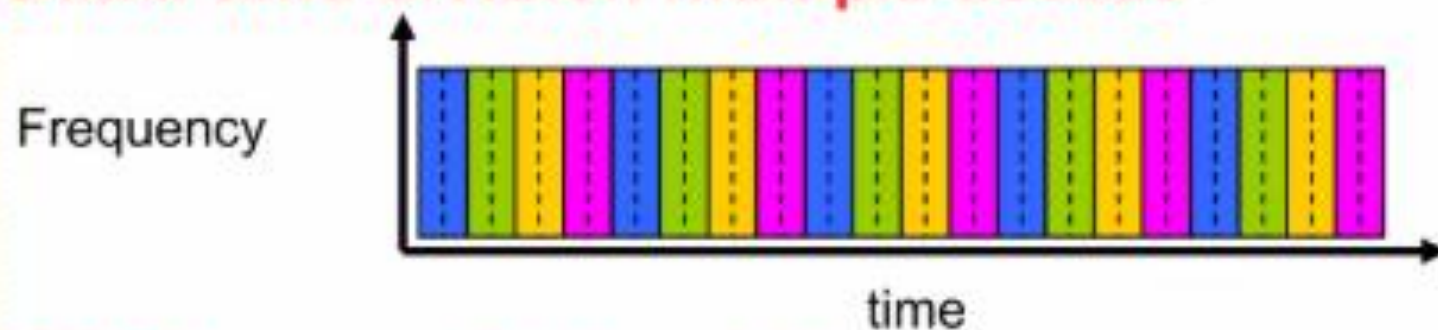
Polling

Token Passing

Figure 12.19 *Select and poll functions in polling access method*



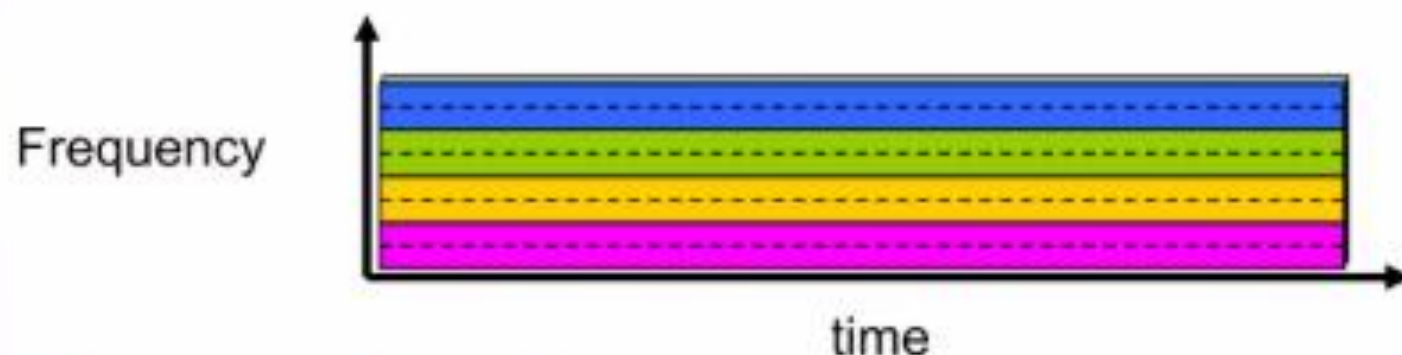
TDMA: time division multiple access



Example:
4 users



FDMA: frequency division multiple access



CDMA: code division multiple access

- Same frequency and time but different codes.



More Questions:

1. Explain the OSI and TCP/IP reference models in detail. Discuss the functions of each layer, comparing how they handle data communication.
2. Describe how different networking devices—hub, switch, repeater, bridge, router, and gateway—operate at various layers of these models.